

OPTIMISATION OF FDM PROCESS PARAMETERS USING TAGUCHI ANALYSIS



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CERTIFICATE OF APPROVAL

(B. Tech Degree in Automobile Engineering)

This project report is hereby approved as a creditable study of an engineering subject carried out and presented in a manner satisfactory to warrant its acceptance as pre-requisite to the degree for which it has been submitted. It is to be understood that by this approval, the undersigned do not necessarily endorse or approve any statement made, opinion expressed and conclusion drawn therein but approve the project report only for the purpose for which it has been submitted.

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ABSTRACT

Fused deposition modeling (FDM) is widely practiced globally in various applications across industries as it is able to create complex 3D shapes and geometries. Industries aim to improve mechanical properties without increasing build time and material consumption and this is a major point of concern as it affects the functionality and cost of the parts made. This challenge can be overcome by cleverly and carefully choosing the correct input parameters. This can directly improve the efficiency of our printed parts.

In this study, effect of varying the input parameters like Slice thickness, Nozzle head speed, Infill density on 9 test pieces made from Nylon 6 are studied. Their effect on mechanical properties like Tensile Strength, Surface roughness, Weight and Build Time is thoroughly studied. Grey Taguchi technique is used to optimize the input parameters. After optimization, a new part was built to further validate our results. The results show that there is a significant improvement in the mechanical properties and the other output parameters considered.

CONTENTS

TOPIC	PAGE NO.
-------	----------

CHAPTER 1

- 1.1. INTRODUCTION
- 1.2. FUSED DEPOSITION MODELING
- 1.3. FDM PROCESS PARAMETRS
- 1.4. FDM 3D PRINTING CHARECTERISTICS

CHAPTER 2

LITERATURE SURVEY

CHAPTER 3

OBJECTIVE OF WORK

CHAPTER 4

TOOLS AND TECHNOLOGY

- 4.1. EXPERIMENTAL DETAILS
- 4.2. Taguchi Method for optimization

CHAPTER 5

PRESENT WORK

- 5.1. Experiment
- 5.2 Result and Discussion

CHAPTER 6

CONCLUSION

Future Scope of Work

REFERENCES

INTRODUCTION

1.1. INTRODUCTION

Additive manufacturing (AM), also known as 3D printing, is a transformative approach to industrial production that enables the creation of lighter, stronger parts and systems. It is yet, another technological advancement made possible by the transition from analog to digital processes. In recent decades, communications, imaging, architecture and engineering have all undergone their own digital revolutions. Now, AM can bring digital flexibility and efficiency to manufacturing operations. Additive manufacturing uses data computer-aided-design (CAD) software or 3D object scanners to direct hardware to deposit material, layer upon layer, in precise geometric shapes. As its name implies, additive manufacturing adds material to create an object. By contrast, when you create an object by traditional means, it is often necessary to remove material through milling, machining, carving, shaping or other means.

In other words, The term “additive manufacturing” references technologies that grow three-dimensional objects one superfine layer at a time. Each successive layer bonds to the preceding layer of melted or partially melted material. Objects are digitally defined by computer-aided-design (CAD) software that is used to create .stl files that essentially "slice" the object into ultra-thin layers. This information guides the path of a nozzle or print head as it precisely deposits material upon the preceding layer. Or, a laser or electron beam selectively melts or partially melts in a bed of powdered material. As materials cool or are cured, they fuse together to form a three-dimensional object. There are seven main additive manufacturing technologies viz Vat photopolymerisation, Material Extrusion, Material Jetting, Binder Jetting, Powder bed fusion, Directed energy deposition, and Sheet lamination. These 3D printing techniques offer different manufacturing styles and can print parts of different materials.

1.2. FUSED DEPOSITION MODELING

The FDM process was introduced commercially to the trade in 1990s as a concept modeling by Stratasys Inc. It is an additive manufacturing technology that is used for a wide array of functions including prototyping, modeling, and production applications [2]. This technology has the capability to create 3D shapes without tooling. Fig. 1 shows the schematic illustration of the FDM process. In FDM process, the feedstock filament material is heated and melted to a semi-liquid state. The semi-liquid material is then extruded through the tip of the nozzle head. The nozzle head moves vertically and horizontally to draw the 3D object directly from a digital data in a layer by layer style. A variety of thermoplastic materials are supported by FDM process, such as ABS, Nylon 12, polycarbonate (PC), and PC-ABS blend.

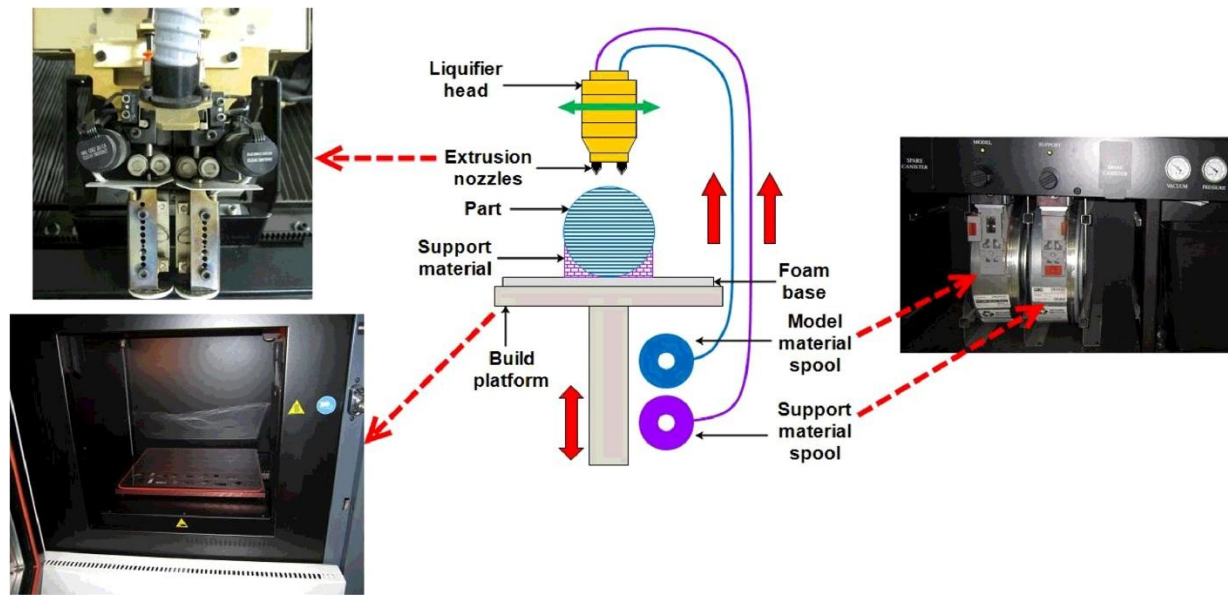


Fig. 1. Schematic representation of the FDM process.

The quality of the manufactured products by 3D printing technologies such as FDM is a crucial factor that determines a company's success within the industry because it sets the appearance, properties, service life, and cost of the final product. Reduction of build time (manufacturing time) and feedstock material consumption without compromising the mechanical performance is the major concern in most industrial applications affecting the manufacturing cost and the functionality of the processed products. Build time can be defined as the amount of time required by the modeler to build the real part on the FDM platform. It is a function of many variables. Compared to other 3D printing processes, the build time in FDM process is considerably dependent on many factors. In general, the FDM is a slower process compared to other 3D printing process such as SLS and SLA from build time point of view. Feedstock material consumption is the amount of material used by the system for building a physical part. Unlike other additive manufacturing processes, such as SLS and SLA, feedstock material consumption and the material deposition rate in FDM are the major factors in reducing build time. The material deposition rate is mainly dependent on some building conditions such as layer thickness and road width. While the mechanical properties of processed parts by FDM are lower than the part manufactured by traditional manufacturing processes. One way to decrease the build time and feedstock material consumption and improve the mechanical performance is through the appropriate adjustments in FDM process parameters to be set for the part. Therefore, the knowledge of the optimum process settings is an important issue in the production and the quality control in the FDM manufacturing process.

The FDM fabrication process works by first loading a spool of thermoplastic filament into the printer. Once the nozzle has reached the desired temperature, the filament is fed into the extrusion head and nozzle, where it melts. The extrusion head is attached to a three-axis system that allows it to move in the x-, y- and z- directions. Melted material is extruded in thin strands and deposited layer by layer in predetermined locations, where it cools and solidifies. Fans can be attached to the extrusion head to accelerate the cooling. To fill an area, multiple passes are required, similar to coloring in a rectangle with a marker. When a layer is finished, the build platform moves down (or in some machine setups, the extrusion head moves up) and a new layer is deposited. This process is repeated until the part is complete.

1.3 FDM PROCESS PARAMETERS

Most FDM systems allow for the adjustment of several process parameters. These include the temperatures of both nozzle and build platform, build speed, layer height and cooling fan speed. An operator usually sets these adjustments, so they are rarely a designer's concern. Build size and layer height are, however, of concern. The build size of a desktop 3D printer is commonly 200 x 200 x 200mm, while industrial machines can be as big as 1,000 x 1,000 x 1,000mm. If a desktop machine is preferred (for example, for cost reduction), a big model can be broken down into smaller parts and then reassembled. Typical layer height used in FDM varies between 50 and 400 microns, and can be determined upon order placement. A layer of shorter height produces smoother parts and captures curved geometries more accurately, while a greater height produces parts more quickly and cost effectively.

1.4 FDM 3D PRINTING CHARACTERISTICS

While there are variations in the extrusion systems of fused deposition modelling 3D printers and parts may have different qualities, there are also some common characteristics (some good, some bad) that you can expect when you FDM print.

Warping

Warping is one of the most common defects in FDM. When extruded material cools during solidification, its dimensions decrease. Since different sections of the print cool at different rates, their dimensions also change at different speeds. Differential cooling causes the build-up of internal stresses that pull the underlying layer upward, causing it to warp, as shown in figure 3. Warping can be prevented through closer temperature monitoring of the FDM system (namely, of build platform and chamber) and increasing adhesion between the part and the build platform.

Layer Adhesion

Secure adhesion between the deposited layers is critical for an FDM part. When the molten thermoplastic is extruded through the nozzle, it is pressed against the previously printed layer. High temperature and pressure cause this layer to remelt and enable the new layer to bond with the previously printed part.

Bond strength between the different layers is always lower than the base strength of the material. This means that FDM parts are inherently anisotropic: their strength in the z-axis is always smaller than their strength in the XY plane. For this reason, it is important to consider part orientation when designing parts for FDM.

For example, tensile test pieces printed horizontally in ABS at 50% infill were compared to test pieces printed vertically and found to have almost four times greater tensile strength in the XY print direction than the Z direction (17.0 megapascals compared to 4.4 megapascals) and elongated almost 10 times more before they broke (4.8% compared to 0.5%). Since the molten material is pressed against the previously printed layer, its shape is deformed to an oval. This means that FDM parts always have a wavy surface, no matter what layer height is used, and that small features, such as small holes or threads, may need to be post-processed after printing.

Support Structure

Because molten thermoplastic cannot be deposited on thin air, some geometries require support structures. Read this detailed article explaining the use of support structure.

Support material might be difficult to remove, so it is often easier to design parts to minimize the need for supports.

Support is usually printed in the same material as the part. Support materials that dissolve in liquid also exist, but they are used mainly in high-end desktop or industrial FDM 3D printers . Using dissolvable supports increases the overall cost of a print.

Infill and Shell Thickness

To reduce print time and save on material, FDM parts are usually not printed solid. Instead, the outer perimeter—called the shell—is traced using several passes, and the interior—called the infill—is filled with an internal low-density structure.

Infill and shell thickness greatly affect the part's strength. For desktop FDM printers, the default setting is 20% infill density and 1mm in shell thickness, which provides a nice compromise of strength and speed for quick prints.

LITERATURE SURVEY

In order to get the best possible mechanical properties and surface finish from a printed part, various studies are conducted on optimizing the various input parameters. Yadav Et al. [1] studied 30 ABS, PETG and multi material test pieces by varying input parameters like infill density, infill pattern, extrusion temperature to achieve the optimum tensile strength which has been further improved by GA-ANN method. Rao Et al. [2] optimized Tensile Strength by choosing layer thickness, print temperature, and infill pattern as the input pattern and concluded that the interactions between layer thickness-infill pattern, and infill pattern-extrusion temperature have significant effect on tensile strength. Rao added that the highest tensile strength of 26.59 MPa is obtained for a layer thickness of 0.1 mm, extrusion temperature of 225 °C and cubic infill pattern. Manoharana Et al. [3] used RSM and Anova to vary layer thickness, Infill density, print speed, temperature and build orientation to optimize tensile strength and the ANN technique gave the minimum value of percentage deviation in Error (1-2%). Naik Et al. [4] studied 3 values each for layer height (0.2, 0.4, and 0.6 mm), nozzle speed (5, 10, and 15 mm s⁻¹), and infill pattern (line, zig-zag and concentric) by using PLA and inferred that Tensile strength increasingly depends on density of the samples. Naik found that the tensile strength of sample with layer height–0.6 mm, nozzle speed–5 mm s⁻¹, and infill pattern—concentric was found to be the highest (54.437 MPa) whereas the extension in sample with layer height–0.4 mm, nozzle speed–15 mm s⁻¹, and infill pattern—line was found to be the highest 13.19%. Nagendra Et al.[5] used Taguchi arrays to improve the surface finish of nylon-aramid composite. The optimum input parameters for Layer thickness, print temperature, raster angle, infill pattern style and infill part density are 0.2mm, 280 °C, 90°, tetrahedral and 90% respectively. Saad Et al. [6] used Particle Swarm Optimization (PSO) to vary the process parameters - layer thickness, printing speed, print temperature and outer shell speed based on the regression model generated from ANOVA.

The flexural strength of PLA from experimental validation obtained using PSO exhibits an improvement of approximately 3.8%. The optimum parameters for layer thickness (*A*), print speed (*B*), print temperature (*C*) and outer shell speed (*D*) of approximately 0.38 mm, 46.58 mm/s, 185.45 °C and 29.59 mm/s result in flexural strength of 96.62 MPa. Ahmad Et al. [7] used Taguchi method and found that surface finish is best with cross print pattern, orientation of 0° on Y-axis, 0° support angle and 0.15mm side walk. Kumar Et al [8] optimized elongation properties of Nylon 6 by varying infill percentage (70, 85 and 100%), bed temperature (50, 60 and 70 °C) and outer perimeters (4, 5 and 6 perimeters) by conducting the analysis based upon the signal to noise (SN) ratio of the elongation property. Mohanty Et al. [9] varied raster angle, part orientation, air gap, layer thickness and raster width to optimize dimensional accuracy in ABS M30 using Taguchi's architecture and recently formulated MARCOS method. The part orientation was found to be the most element. Ramesh Et al. [10], optimized the Ultimate Tensile Strength (UTS), impact strength, flexural strength and shore D hardness of Nylon by varying fill Density (FD) , Layer Height (LH) and Print Speed (PS) using Taguchi L9 orthogonal array and ANOVA. Kumar Et al. [11] also optimized multiple Mechanical properties like tensile, flexural and hardness of Carbon Fibre Reinforced PETG by using different print speeds, infill densities and layer heights. Using Taguchi L9 and ANOVA the optimum tensile, flexural strength and hardness obtained in this research are 31.567 MPa, 35.045 MPa and 67.0011 BHN respectively.

The most desirable machine parameters that have come out in this research are print speed 60 mm/sec, infill density 80% and layer height 200 μ . Kamoona Et al. [12] studied the result of varying air gap, raster angle, and build orientation on the impact strength and mechanical properties using Response Surface Methodology (RSM) based on face centered central composite design, ANOVA. Their study reveal that the part build Y-orientations (flat) at 0° and 45° have a significant directly proportional influence on the impact strength, while Z-orientation (upright) at 90° has indirectly proportional effect on the impact strength. Moreover, raster angle has a much significant directly proportional influence on the impact strength at 0° and 60° angles and indirectly proportion influence at 30°. Nagendra, Et al. [13] made another study to enhance multiple mechanical properties such as tensile strength, flexural strength, impact strength and compression strength of Nylon, Aramid compounds by the Gray Taguchi technique, Analysis of variance using Minitab 18 software.

The Layer thickness, print temperature, raster angle, infill part density and infill pattern style were chosen as the input parameters. The study revealed that layer thickness of 0.4 mm, print temperature of 300 °C, infill part density of 90%, raster angle of 90° and infill pattern style of rectilinear were the optimum input. Giri Et al. [14] used function approximation of Artificial Neural Network (ANN) to optimize tensile strength surface roughness, and build time of PLA. Their study found that ANN can predict the experimental data with a coefficient of correlation $R = 0.9981, 0.9984, 0.99837$ subsequently for tensile strength, Build time, and surface roughness and root mean square error as 0.5543, 0.578 and 0.241 for three outputs. They also concluded that among the input parameters - layer thickness, air gap, raster width, build orientation, raster angle, and the number of contours, the build orientation is the most important parameter for optimum results. Srivastava, Et al. [15] chose contour width, air gap, raster angle, and spatial orientation to optimize model material. The optimum process parameter conditions were obtained from grey relational grade. The results obtained provide useful information of the method to control responses and ensure minimal build time and model material volume for prototyping requirements. The assessment outcome provided a scientific reference to obtain minimal values of build time and model material volume utilized, and it was found out that these correspond to a contour width of 0.654 mm, air gap of 0.0254 mm, raster angle of 0°, and orientation of 0°. Moradi, Et al. [16] investigated layer thickness (0.15 mm-0.35 mm), infill percentage (15-55%), and the number of contours (2-6) to optimize maximum failure load, parts weight, elongation at break, and build time of nylon using ANOVA.

Based on the results, the layer thickness is the significant primary variable for all responses. The experimental evaluation showed that the maximum values of failure load and elongation were obtained by changing the layer thickness from the lowest to the highest. By reduction in layer thickness it was found that, at the same printing speed, the cooling rate increases, which results in greater strength and less elongation. As a result, it could be concluded that by increasing the number of contour layers from 2 to 6, the maximum failure force increased 42%. Increasing the contours due to the similar effect to increasing the infill density, increases the failure force and production time.

After doing a detailed literature review, it was found that a lot of work has already been done to enhance the Tensile Strength [1-4], Surface roughness [5,7], and mechanical properties like flexural strength [6], elongation [8], dimensional accuracy [10]. But, in these studies the input parameters were chosen to obtain a single optimum output parameter. Moreover, Studies on multiple mechanical properties [10-13] fail to acknowledge the effect of varying the input parameters on other properties which in real life usage may diminish the quality of the properties not considered. Similarly, studies that have optimum parameter settings for surface roughness may yield unsuitable mechanical properties or may take more time to build the part.

Studies that have considered multiple contrasting properties like Mechanical properties, surface roughness and build time [14-15] failed to consider infill density as a significant input parameter. Moreover, only a very limited amount of work is done on Nylon 6. Studies like that done by Moradi, Et al. [16] have not optimized surface roughness which is an important output parameter. In order to fill the above research gaps, this study contributes to the literature in many ways. As most of the earlier studies failed to consider optimizing multiple output parameters, this study considers four contrasting output characteristics namely Mechanical properties like Tensile strength and Elongation, surface roughness, weight of parts and build time by considering Layer thickness, Infill density and Nozzle velocity as the input characteristics.

OBJECTIVE OF THE WORK

FDM process has a huge application in automotive manufacturing industries. It is capable of producing complex light weight parts in record time. Producing parts that have good mechanical properties in small amount of time is however a challenge. Engineers have often struggled to find the optimum set of input characters to achieve contrasting output characteristics]of achieving good mechanical properties like tensile strength or surface roughness in small time. Also the weight of the built parts are crucial. In the present work, input characters like Nozzle head speed, Slice thickness, Infill density of Nylon 6 is optimized using Grey Taguchi technique. The input parameters are optimized in order to achieve the best possible Tensile strength, Surface roughness, Build time and Weight of the build parts. The optimum process setting is achieved and a new part is built to validate the experimental results.

CHAPTER 4

TOOLS AND TECHNOLOGY

4.1. Experimental Details

During experimentation Nylon 6, an engineering derived thermoplastic material was used due to its excellent mechanical properties such as toughness, elasticity, chemical resistivity, high melting points, and moldability.

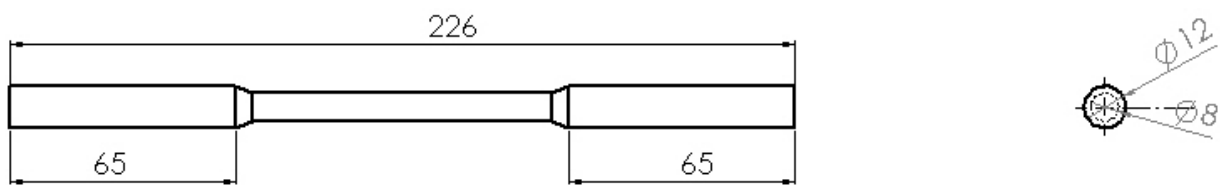


Fig Schematic diagram of specimen

The CAD files were converted into .stl file by the Ultimaker Cura software. In the software we can provide different values for various process parameters such as Head speed, infill density, slice thickness etc. After selecting the parameters, the .stl file is converted into a G-code file that can be read by the printer.

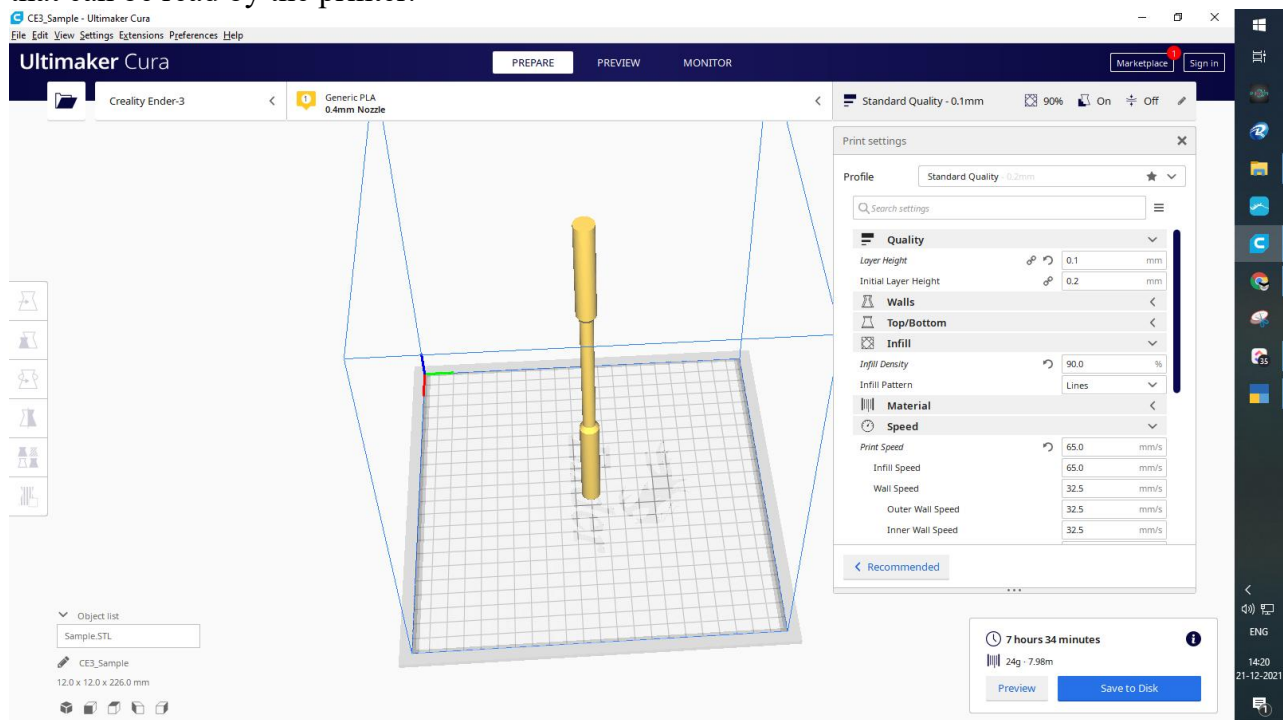


Fig: User interface of Ultimaker Cura

4.2 Taguchi Method for optimization

Classical experimental design methods are time consuming and require more experiments to be performed when the number of control factors is high. Taguchi methods¹⁰ use a special design of orthogonal arrays to study the entire factor with only a small number of experiments. In the Taguchi method, the term ‘signal’ represents the desirable value for the output characteristic and the term ‘noise’ represents the undesirable value for the output characteristic. Therefore, the term ‘signal-to-noise ratio’ or S/N ratio in Taguchi method measures the deviation of quality characteristic from the desired value. Optimization in Taguchi method means maximization of S/N ratio. In the present problem for simultaneous optimization of kerf quality and kerf deviation following steps of Taguchi method has been followed:

- i) *Computation of quality loss function*: Quality loss functions which may be of three types based on output characteristics: smaller-the-better, nominal-the-best or higher-the-better [montgomery]. As in the present problem minimum value of kerf width and kerf deviation is desired, the smaller-the-better quality loss function (f) has been used and for i th trial ($\hat{f}_i$) it is given

$$\text{by: } f_i = \frac{1}{n} \sum_{i=1}^n y_i^2 \quad (1)$$

where, y_i are the output or quality characteristics at the i th trial of same parameter level, and n is the number of trials.

- ii) *Normalization of quality loss function*: As Present work bears two quality characteristic (kerf width and kerf deviation) with different range of values, it is required to normalize the quality loss function values computed in Eq.(1) between 0 and 1 before employing them for simultaneous optimization.

Normalised quality loss values are computed as $\hat{f}_{ij} = \frac{f_{ij}}{f_{i\max}}$ (2)

Where, $\hat{f}_{ij}$ = normalized quality loss for i th output characteristic at the j th trial condition,

$f_{i\max}$ = maximum quality loss for the i th output characteristic among all the trials/experimental runs. If W_i represents the weights corresponding to i th output characteristic, p is the number of output characteristics, then the total normalised quality loss for j th trial condition (F_j) can be computed

as $F_j = \sum_{i=1}^p w_i \hat{f}_{ij}$ (3)

- iii) *Computation of multiple S/N ratio*: Multiple S/N ratio (MSNR) is computed to determine the quality index of each design point using total normalised quality loss values. The MSNR corresponding to j th trial condition (MSNR_j) is calculated as:

$$MSNR_j = -10 \log_{10}(F_j) \quad (4)$$

- iv) *Optimization of process parameters*: Mean of all MSNR values at a specific level of a process parameter is used to describe the effect of a process parameter or factor on quality characteristics at that level. A parameter level corresponding to the maximum average MSNR is called the optimum level for that parameter. The predicted value of MSNR (MSNR_{opt}) at optimum parameter levels is calculated by using following formula:

$$MSNR_{opt} = \overline{MSNR} + \sum_{i=1}^k (MSNR_{mi} - \overline{MSNR}) \quad (5)$$

where $\overline{MSNR}$ is the mean MSNR of all experimental runs, k is the number of significant control factors, and m_i is the average MSNR for i th control factor corresponding to optimum parameter level. Further an experiment is conducted with optimum parameter levels determined by Taguchi method for validation of the predicted response.

PRESENT WORK

5.1 Experiment

In the present investigation, experiment has been carried out using Arya_Uno+ FDM type 3D printer, printed by 0.4mm nozzle in Nylon Filament. As Nylon 6 is hydrostatic, the 3d printer is fully enclosed type



Arya_Uno+ FDM type 3D printer

The ASTM standard printed specimen has been printed using the following initial input parameters: Slice thickness- 0.1mm, Head speed- 25 m/min, Infill density- 20%. Three levels of each input process parameters have been selected. Range of operations for process parameters are 0.1- 0.3 mm for Slice thickness (ST), 25-65 m/min for Head speed (HS), 20-90 % for Infill density (ID) respectively. The numerical value of control factors at different levels for Nylon 6 is given below.



Nylon 6 Specimen

Symbol	Factors	Levels		
		1	2	3
A	Slice Thickness (mm), ST	0.1	0.2	0.3
B	Head Speed (m/min), HS	25	45	65
C	Infill Desity (%), ID	20	50	90

Table 1: Levels of experimental input parameters

Nine experiments have been conducted based on L9 orthogonal array experimental design as given in Table 2. Time is measured using stopwatch and it is representative of the total time required to build the part i.e from printing layer1 to the final layer.

Exp. No	Coded Input parameters			Output parameter
	Slice Thickness	Head Speed	Infill Desity	Time
	ST	HS	ID	T
	(mm)	(m/min)	(%)	(min)
1	1	1	1	154
2	1	2	2	220
3	1	3	3	293
4	2	1	2	111
5	2	2	3	148
6	2	3	1	84
7	3	1	3	101
8	3	2	1	59
9	3	3	2	77

Table2: Experimental dataset for Nylon 6 using L9 orthogonal array

5.2 Results and discussion

The experimental dataset developed based on L9 orthogonal array with Time as output is given in Table 2. The quality loss values for quality characteristic T during each experimental run is calculated using Eq.1. The quality loss, normalised quality loss and SNR values are computed using Eq.2-4 and given in Table 3.

Exp No.	Quality loss	Normalised Quality loss	SNR(dB)
1	23716	0.2763	5.5869
2	48400	0.5638	2.4889
3	85849	1.0000	0.0000
4	12321	0.1435	8.4309
5	21904	0.2551	5.9321
6	7056	0.0822	10.8518
7	10201	0.1188	9.2509
8	3481	0.0405	13.9203
9	5929	0.0691	11.6075
Mean of SNR (η)			7.5633

Table3: Quality loss values and multiple S/N ratio for Taguchi method

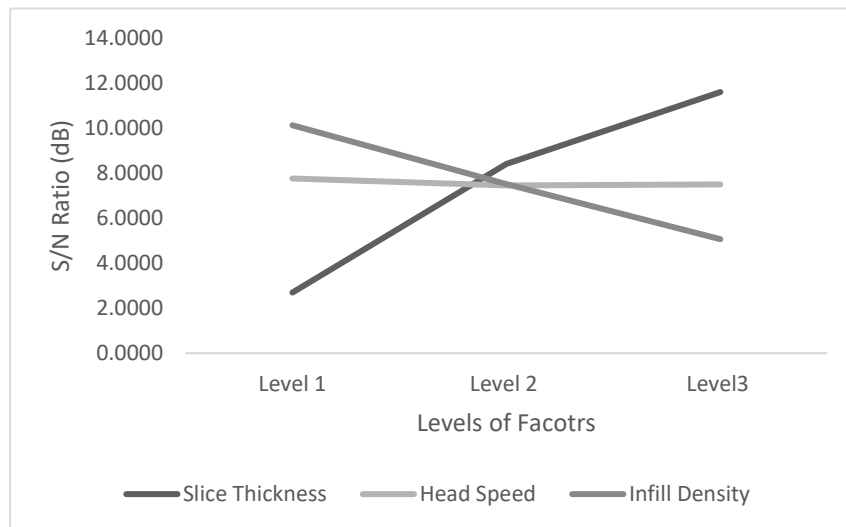
The effect of different operating parameters on S/N ratio comprising Time is shown in Table 4 and the Fig below. It is clear that, optimum levels of different control factors for obtaining Time are: Slice Thickness at level 3 (0.3mm), Head Speed at level 1(25 m/min) and Infill density at level 1 (20%) Relative contribution of the controlling parameters on Time is shown in Table 4.

Symbol	Factors	Mean of S/N Ratios (dB)			Maximum - Minimum	Rank
		Level 1	Level 2	Level3		
A	Slice Thickness (mm), ST	2.6919	8.4049	11.5929 ^a	8.9010	1
B	Head Speed (m/min), HS	7.7563 ^a	7.4471	7.4864	0.3091	3
C	Infill Desity (%), ID	10.1197 ^a	7.5091	5.0610	5.0587	2

^a optimum level

s

Table4: Effect of factor on multiple S/N ratio



Effect of factor levels on S/N ratio

	Initial Parameter level	Optimal Input parameters
		Prediction
Level	A1B1C1	A3B1C1
Time	154	-
SNR (dB)	5.5869	14.3423
Improvement of SNR		8.7554

Table5: Results of validation experiment at optimum parameter level

Predicted value of SNR with optimized parameter levels is computed using Eq.5 and given in Table 5. The validation experiment has been conducted with above mentioned optimum levels of operating parameters and optimized output with corresponding SNR is given in Table5. The significant improvement in SNR is observed to be 8.7554 dB during comparison with initial parameter setting.

CONCLUSION

In present work, multi-objective optimisation of Build Time has been carried out using Taguchi method for FDM 3D Printing of Nylon 6. Following conclusions can be drawn on the basis of results obtained:

- *Results of Taguchi optimization indicates that minimum build time can be achieved by operating with input parameter setting of slice thickness is put to level 3, Head speed is put to level 1 and infill density is put to level 1.*
- *Variation of Head speed is insignificant across the 3 levels even if our result yields level 1 as optimum.*
- *The S/N ratio compared to initial conditions has been improved by 8.7554 dB.*

FUTURE SCOPE OF WORK

We had initially planned to optimize four output parameters i.e Tensile Strength, Surface roughness, weight and build time. But due to the closure of our institution as the situation outside worsened, we were unable to perform the Tensile testing and Surface roughness testing of the 9 printed specimen. But we will continue the work as soon as the situation permits and we will submit a fresh report post that.

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